AHMED NOORIM

B.Sc. in Computer Science and Engineering

Email: ahmed.noorim07@gmail.com
Location: Chattogram, Bangladesh
Contact: +880 1743 980099

LinkedIn: linkedin.com/in/ahmed-noorim
Google Scholar: AhmedNoorim (click)
Portfolio: AhmedNoorim (click)

EDUCATION

May 2021 - Jan 2025 **B.Sc. in Computer Science and Engineering**, East Delta University

- CGPA: **3.91 out of 4.00** ***(summa cum laude)***
- Thesis: *An Ensemble Deep Learning Framework for Two-Stage Diabetic Retinopathy Diagnosis and Severity Screening.*

RESEARCH INTERESTS

- Computer Vision
- Biomedical Informatics
- Internet of Things (IoT)
- Medical Image Analysis
- Multimodal Models
- Edge/Cloud Computing
- Trustworthy & Explainable AI
- Connected Autonomous Vehicles

RESEARCH EXPERIENCE

Jun 2024 - Present **Research Assistant**, Research Activator Lab

- Co-authored two IEEE conference papers: NCIM 2025, BECITHCON 2024
- Trained and fine-tuned deep learning models (YOLOv11, U-Net, Ensemble CNN).
- Performed structured metadata extraction and screening for literature review.

Jun 2026 - Present **Systematic Review (in progress)**, Explainable Multimodal LLMs for Medical Report Generation

- 91 studies screened from Q1 venues (IEEE, Nature, Elsevier, ACM, Springer, Frontiers).
- Building a taxonomy of explainability methods and evaluation metrics.

Jul 2024 - Jan 2025 **Undergraduate Thesis Research**, East Delta University

- Designed a two-stage diabetic retinopathy (DR) framework: a DenseNet121 detector to detect DR, followed by a weighted ensemble (DenseNet121, Xception, EfficientNetB1) for four-grade DR severity grading.
- Merged APTOS 2019, MESSIDOR-2, and IDRiD into a 5,922-image dataset; benchmarked three fundus image preprocessing pipelines.
- Published at the International Conference on Information and Communication Technology for Intelligent Systems 2025 as first author.

CONFERENCE PROCEEDINGS

1. **Noorim, A.**, Jahan, R. N., Ahsan, M. S. A., Adhikary, S., & Uddin, M. J. (2025). An ensemble deep learning Framework for Two-Stage Diabetic Retinopathy Diagnosis and Severity Screening. Smart Innovation, Systems and Technologies, 333–342. https://doi.org/10.1007/978-981-95-1365-9_27
2. Uddin, A., Ananna, F. B. F., Farah, N. T., **Noorim, A.**, Hosen, M., Roy, P. S., & Tasnia, N. (2025). PotholeNet11:A YOLO11-Powered Real-Time framework for comprehensive pothole detection and road safety. International Conference on Next-Generation Computing, IoT and Machine Learning (NCIM), 1–6. https://doi.org/10.1109/ncim65934.2025.11160097
3. Farah, N. T., Pushpita, S. H., **Noorim, A.**, Saha, A., Hosen, M., & Uddin, A. (2024). Enhancing Lyme Disease Erythema migrans rashes Classification: U-Net Segmentation and Ensemble Deep Learning Model. IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON), 218–223. https://doi.org/10.1109/becithcon64160.2024.10962629

SELECTED PROJECTS

1. **Bengali Handwritten Alphabet Classification Web App**. (View)
    a. Assembled a dataset of about 20,000 images, combining self-made and internet-sourced images.
    b. Trained a CNN from scratch in TensorFlow and deployed it via a Flask API and Web Interface.
2. **Vision Based Automated Attendance System.** (View)
    a. Designed a system that automates QR code scanning and attendance recording for an institution.
    b. Configured an ESP32-CAM with an FTDI module in the Arduino IDE to capture and scan QR codes.
    c. Implemented backend logic in Python, database in MongoDB, and dashboard via Flask.

TEACHING & MENTORING

- Led peer study groups at university, organising concept-clearing sessions for 8-12 students per term.
- Mentored junior members of the Research Activator Lab on literature search and academic writing.

TECHNICAL SKILLS

- **Languages:** Python, C++, Arduino
- **ML/DL:** TensorFlow, PyTorch, OpenCV, scikit-learn, Keras
- **Tools:** Flask, MongoDB, Kaggle, Anaconda, LaTeX, VS Code
- **Research Methods:** Systematic literature search, metadata extraction, academic writing

HONORS & AWARDS

- Board of Trustees Grant Scholarship (merit-based, competitive), East Delta University, 2021-2025
- *Summa cum laude*, B.Sc. in CSE, East Delta University, 2025
- 2nd Runner-Up (Poster Presentation), East Delta University, 2025
- Vice Chancellor's List and Dean's List, East Delta University

LEADERSHIP & SERVICE

- **Vice Convener of East Delta University Erudition Club** May 2022 - Apr 2024
    - Organized academic talks and inter-departmental events for 100+ members.
- **Documentation Secretary of ShokherMulluk** Apr 2018 - Aug 2021
    - Maintained records and reports for a community organisation.
- **Volunteer at Hope Foundation** Jan 2024 - Jun 2025
    - Arranged educational and celebratory festival events for rural children.

STANDARDIZED ENGLISH TEST

January 2026	**IELTS Overall 7.0** [Listening 8.0, Reading 7.0, Writing 6.5, Speaking 7.0]
October 2025	**GRE Overall 302** [Quant 161, Verbal 141, Writing 3.5]

REFERENCES

Md. Ishtiaque Aziz Zahed, PhD
Associate Professor, Academic Mentor
School of Science, Engineering, and Technology
East Delta University
ishtiaque@eastdelta.edu.bd

Md. Sabbir Al Ahsan
Lecturer, Thesis Supervisor
Department of CSE
East Delta University
sabbir.a@eastdelta.edu.bd

Arshiana Shamer
Lecturer, Project Supervisor
Department of CSE
East Delta University
arshiana.s@eastdelta.edu.bd